HYPERRIGIDITY IN LOW DIMENSIONS AND A DIMENSION-33 COUNTEREXAMPLE

Abstract. We study the least affine dimension of a commutative counterexample to Arveson's hyperrigidity conjecture. A quadratic estimate for supporting affine functions relates the unique extension property to inverse-square energy. Using the normal lift, we prove hyperrigidity for three-dimensional convex bodies with closed extreme boundary when every extreme point is exposed or when the boundary is C^1. We also modify the Walsh dilation of Bilich–Scherer to construct a counterexample in affine dimension 33, with a 34-dimensional function system and a diffuse representation. Finally, we establish two obstructions to smaller versions of that construction. For the stationary model considered here, arbitrary vector directions do not yield its quadratic-distance certificate in horizontal dimension at most four; more generally, no finite strictly convex graph can repair the stationary construction in affine dimension four. The latter conclusion requires no smoothness. The least affine dimension remains between 3 and 33.

1. Introduction

Let $K \subset \mathbb{R}^d$ be a compact convex set with closed extreme boundary $X = \operatorname{ex} K$. We write $A(K)$ for the complex vector space of continuous affine functions on K and regard

$$A(K)|_X \subset C(X)$$

as a function system. A unital representation $\pi : C(X) \to B(H)$ has the *unique extension property* (UEP) if the only unital completely positive (UCP) map $\Phi : C(X) \to B(H)$ agreeing with π on $A(K)|_X$ is π itself. By Arveson's characterization, the function system is hyperrigid exactly when every unital representation has this property [Arv]. We use this formulation throughout.

Each point evaluation at an extreme point has the unique extension property: a probability measure with that barycenter is a point mass. The question is whether the same uniqueness survives for diffuse representations. Bilich and Scherer showed that it need not [BS]. Their construction raises a geometric question: how small can the affine dimension of a commutative counterexample be?

The present paper gives positive results in dimension three, an explicit counterexample in dimension 33, and structural restrictions on attempts to bridge the gap. Its first main result is the following.

Theorem A. *Let $K \subset \mathbb{R}^3$ be a compact convex body with closed extreme boundary. Then $A(K)|_{\operatorname{ex} K}$ is hyperrigid if either*

(i) *every extreme point of K is exposed; or*
(ii) *∂K is C^1.*

In particular, every strictly convex three-dimensional body is covered, without any smoothness assumption.

The two hypotheses have different roles. Exposedness supplies an affine function separating each represented point from every other point. Smoothness instead supplies a unique normal,

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which allows a reduction to planar supporting faces. Neither hypothesis is assumed for general three-dimensional convex bodies.

The common mechanism is an energy estimate. If n_x and n_y are unit outward normals at $x, y \in \partial K$, then

$$0 \le \langle n_x, x-y\rangle \le \frac{1}{4}\big|(x+n_x)-(y+n_y)\big|^2. \tag{1.1}$$

The normal lift $x \mapsto x+n_x$ takes values on $\partial(K+\overline{B})$, where $\overline{B}$ is the closed unit ball. In dimension three, every nonzero measure on this parallel surface has infinite inverse-square energy. On the other hand, a nonzero separated block of an agreeing positive operator-valued measure would produce a finite-energy measure. This contradiction proves the required separation. For a smooth boundary, we first obtain intertwining over the normal space and only then apply planar hyperrigidity to the supporting faces. The complete statements and proofs are given in Theorems 4.4 and 5.1.

Our second main result is constructive.

Theorem B. *There is a compact convex body $K \subset \mathbb{R}^{33}$ with closed extreme boundary X, a unital representation $\pi : C(X) \to B(H)$, and a unital completely positive map $\Phi : C(X) \to B(H)$ such that*

$$\Phi|_{A(K)|_X} = \pi|_{A(K)|_X}, \qquad \Phi \neq \pi.$$

The Hilbert space H is separable and infinite dimensional, and the spectral measure of π has no point atoms. Moreover $\dim_{\mathbb{C}} A(K) = 34$.

The construction is a change of distribution in the Walsh dilation of Bilich–Scherer, rather than a new dilation mechanism. Theorem 6.1 gives the complete data. The operator identities are proved before the numerical parameters are chosen; the remaining containment estimate is a scalar inequality for every infinite digit sequence. A continuous barycentric kernel then gives the extension, and a quadratic function provides an explicit positive witness of disagreement.

For comparison, [BS, Corollary 3.3] proves a construction in ambient dimension 129, states an optimization to dimension 57, and mentions an unverified dimension-11 proposal in Remark 3.4. Scherer's planar theorem [Sch1, Theorem 3.8] excludes dimensions at most two. Theorem B therefore gives

$$3 \le d_{\min} \le 33, \tag{1.2}$$

where $d_{\min}$ denotes the least affine dimension under the closed-extreme-boundary hypothesis. The conjectural value four is not established here. The positive theorem for compact spectrahedra [Sch2, Theorem 3.7] is a further point of comparison; it applies to finite self-adjoint affine matrix pencils, not arbitrary projections of spectrahedra.

The construction obstructions explain why simply changing its visible geometry does not close this gap. A finite-dimensional signing lemma rules out the quadratic-distance certificate for every stationary three-point distribution and every geometric-norm vector encoding in horizontal dimension at most four, under the nonnegative-potential and exact-height assumptions of Theorem 7.2. Thus even arbitrary directions cannot make that certificate work in affine dimension four or five. A stronger geometric argument rules out *every finite strictly convex graph* for the same stationary mechanism in affine dimension four (Corollary 8.6). It uses positive distance levels of the represented set and requires no differentiability of the graph. These are obstructions to a specified construction family, not universal lower bounds on $d_{\min}$.

The general supporting-gap criterion, Theorem 3.1, is useful beyond these applications. Appendix A gives a source in a four-dimensional convex body whose normal lift supports a finite-energy measure, but whose representations still have the unique extension property by a

sharper metric. Appendices B and C treat varying and correlated distributions. Appendix D records two restrictions on direct operator constructions. Exploratory numerical searches and superseded certificates are not needed for any proof in this paper.

2. Representations, measures, and defects

All Hilbert spaces are complex. Representations and completely positive maps are unital unless stated otherwise. A convex body has nonempty interior in its ambient Euclidean space. We write $\overline{B}$ for the closed unit ball and use $|\cdot|$ for Euclidean norms. For a compact convex set K, its support function is $h_K(u) = \max_{x\in K}\langle u, x\rangle$. An exposed point is the unique minimizer of an affine function on K. A smooth boundary point has a unique unit outward normal. A strictly convex body has no nondegenerate segment in its boundary; every boundary point is then exposed.

Whenever $X = \operatorname{ex} K$ is compact, the coordinate functions and the constant function generate $C(X)$. If $\operatorname{aff} K = \mathbb{R}^d$, their complex linear span is $A(K)|_X$ and has dimension $d+1$. The dimension of a representation space is a separate issue.

For a representation $\pi : C(X) \to B(H)$ we write P for its spectral measure. A UCP map Φ has a positive operator-valued measure E. One may construct its bounded Borel extension from a Stinespring representation $\Phi(f) = V^*\rho(f)V$: if Q is the spectral measure of ρ, set $E(B) = V^*Q(B)V$. All operator-valued measures below are countably additive in the strong operator topology. Saying that a representation is concentrated on a Borel set Z means $P(Z) = I$; this is stronger than an assertion about the closure of that set.

Lemma 2.1 (Coordinate defects)**.** *Suppose $\Phi = V^*\rho(\,\cdot\,)V$ agrees with π on the affine coordinates $x_1, \dots, x_d$. Then*

$$D_j := \Phi(x_j^2) - \pi(x_j)^2 = \big[(I - VV^*)\rho(x_j)V\big]^*\big[(I - VV^*)\rho(x_j)V\big] \geq 0.$$

All D_j vanish if and only if VH reduces $\rho(C(X))$, and this is equivalent to $\Phi = \pi$.

Proof. Expand the square and use $V^*\rho(x_j)V = \pi(x_j)$. If it vanishes, VH is invariant for the self-adjoint operator $\rho(x_j)$ and hence reduces it. The coordinates generate $C(X)$, so the compression is a representation agreeing with π on a generating set. The converse follows from the same identity. □

Lemma 2.2 (Atomic representations)**.** *Every Hilbert direct sum of point evaluations at points of X, with arbitrary multiplicities, has the unique extension property.*

Proof. Write $H = \bigoplus_x H_x$, with $\pi(f)|_{H_x} = f(x)I$. For a unit vector $h \in H_x$, the scalar measure representing $f \mapsto \langle \Phi(f)h, h\rangle$ has barycenter x and is therefore δ_x. Indeed, a probability measure with an extreme barycenter cannot be split into two parts with different conditional barycenters. In a Stinespring representation this gives

$$\|\rho(f)Vh - f(x)Vh\|^2 = \langle \Phi(|f - f(x)|^2)h, h\rangle = 0.$$

Thus $\rho(f)V = V\pi(f)$ on each summand and then on their Hilbert direct sum. This proves equality, including all off-diagonal coefficients. □

In particular, finite-dimensional representations cannot witness a counterexample. We shall also need a stronger local consequence of extremality.

Lemma 2.3 (Local vanishing of the defect)**.** *Let Φ agree with π on $A(K)|_X$, and let D_j be the defects above. If x belongs to the spectral support of π, then*

$$\lim_{r\downarrow 0} \left\| P(B(x,r)) \left(\sum_j D_j \right) P(B(x,r)) \right\| = 0.$$

Consequently no coordinate defect, nor their sum, is bounded below by a positive multiple of the identity.

Proof. Any sequence of probability measures on X whose barycenters tend to x converges weakly to δ_x: every weak limit has the extreme barycenter x. By compactness, their second moments about x tend to zero uniformly as their barycenters approach x. For a unit vector $h \in P(B(x,r))H$, the scalar measure of Φ at h has barycenter within r of x. Affine agreement gives

$$\left\langle \sum_j D_j h, h \right\rangle = \int_X |y-x|^2 \, d\mu_h(y) - \left\langle \pi(|\,\cdot - x|^2)h, h \right\rangle \le \int_X |y-x|^2 \, d\mu_h(y).$$

The right side tends to zero uniformly in these unit vectors. Positivity gives the norm estimate. Every such spectral neighborhood is nonzero, which proves the last assertion. □

3. A quadratic affine-barrier criterion

The positive results in this section rest on a common observation. An affine supporting function can bound the amount of a dilation that passes from one part of the spectrum to another. If these bounds vanish quadratically in a suitable metric, a nonzero off-diagonal effect would produce a scalar measure of finite inverse-square energy. A critical covering estimate rules out such a measure. The metric need not be the ambient Euclidean metric, and this flexibility will be useful below.

We retain the notation of Section 2: P is the spectral measure of π, and E is the POVM of an affine-agreeing UCP map Φ. Their bounded Borel extensions will be used without changing notation.

Theorem 3.1 (Quadratic affine barriers)**.** *Let $\pi : C(X) \to B(H)$ be a unital representation, and let $Z \subset X$ be Borel with $P(Z) = I$. Suppose that, for each $x \in Z$, an affine function b_x on K is specified so that its coefficients depend Borel measurably on x and*

$$b_x(x) = 0, \qquad b_x(y) > 0 \quad (y \in K \setminus \{x\}).$$

Suppose also that there are a compact metric space (T, d_T), a Borel map $\theta : Z \to T$, and a finite constant M such that

$$b_x(y) \le M \, d_T(\theta(x), \theta(y))^2 \qquad (x, y \in Z).$$

Assume that every nonzero finite positive Borel measure μ on Z satisfies

$$\iint_{Z\times Z} d_T(\theta(x), \theta(y))^{-2} \, d\mu(x) d\mu(y) = +\infty,$$

where the integrand is $+\infty$ when the distance is zero. Then π has the unique extension property relative to $A(K)|_X$.

We first establish the analytic facts needed for the proof. For a finite positive measure ν on a metric space, write

$$I_2(\nu) = \iint d(u,v)^{-2} \, d\nu(u) d\nu(v).$$

A Borel set has *zero inverse-square capacity* here if every nonzero finite positive measure concentrated on it has infinite I_2-energy. The formulation in Theorem 3.1 is the corresponding statement for the pushforwards under θ; it does not require a separate measurability assertion about the image of θ.

3.1. **Energy and localization.**

Lemma 3.2 (Rank-one domination)**.** *Let P be a spectral measure on a compact metric space, let w be a bounded nonnegative Borel function, and suppose*

$$|\eta\rangle\langle\eta| \le M \int w\, dP.$$

Then, for $\mu_\eta(D) = \langle P(D)\eta, \eta\rangle$,

$$\int w^{-1}\, d\mu_\eta \le M,$$

with $w^{-1} = +\infty$ on the zero set of w.

Proof. Put $W = \int w\, dP$. For $\epsilon > 0$, set $h = (W + \epsilon I)^{-1}\eta$ and $t = \langle \eta, h\rangle \ge 0$. The assumed domination gives

$$t^2 \le M\langle Wh, h\rangle \le M\langle (W + \epsilon I)h, h\rangle = Mt.$$

Thus $t \le M$. The spectral theorem and monotone convergence as $\epsilon \downarrow 0$ prove the assertion. □

Lemma 3.3 (Critical covering bound)**.** *Let T be a compact metric space. Suppose that, for every sufficiently small $\epsilon > 0$, it can be covered by at most $M\epsilon^{-2}$ sets of diameter at most ϵ. Then $I_2(\nu) = +\infty$ for every nonzero finite positive Borel measure ν on T. In particular, this conclusion holds for the boundary of every compact convex body in $\mathbb{R}^3$.*

Proof. The covering sets may be replaced by their closures and then refined to a Borel partition $T = \bigsqcup_{j=1}^{N_\epsilon} T_j$, with $N_\epsilon \le M\epsilon^{-2}$ and $\operatorname{diam} T_j \le \epsilon$. If $m = \nu(T) > 0$, Cauchy–Schwarz gives

$$(\nu \times \nu)\{d(u, v) \le \epsilon\} \ge \sum_j \nu(T_j)^2 \ge \frac{m^2}{N_\epsilon} \ge \frac{m^2}{M}\epsilon^2.$$

The layer-cake formula and Tonelli's theorem therefore imply

$$I_2(\nu) \ge 2\int_0^{\epsilon_0} t^{-3}(\nu \times \nu)\{d(u, v) \le t\}\, dt \ge \frac{2m^2}{M}\int_0^{\epsilon_0} \frac{dt}{t} = +\infty.$$

For the final assertion, translate the convex body so that $r\overline{B}_3 \subset K \subset R\overline{B}_3$ for some $r, R > 0$. Its Minkowski functional j_K is $1/r$-Lipschitz and is at least $1/R$ on S^2. Hence the radial map $u \mapsto u/j_K(u)$ is Lipschitz from S^2 onto ∂K. The sphere admits the required $O(\epsilon^{-2})$ covers, as follows from finitely many Lipschitz coordinate charts. The radial map transfers this covering bound to ∂K. □

The next lemma isolates the operator argument. Unlike Theorem 3.1, it does not require each affine function to expose a single point.

Lemma 3.4 (Local annihilation)**.** *Suppose Φ and π agree on $A(K)|_X$. Let $F, G \subset X$ be Borel. For each $x \in F$, let b_x be an affine function nonnegative on K. Suppose that*

$$b_x(y) \ge \delta > 0 \quad (x \in F,\ y \in G), \qquad b_x(y) \le M\, d_T(\theta(x), \theta(y))^2 \quad (x, y \in F),$$

where $\theta : F \to T$ is Borel and T is compact metric. If every nonzero finite positive measure on F has infinite energy in the displayed metric, then $E(G)P(F) = 0$.

Proof. Set $A = P(F)E(G)P(F)$. For each fixed $x \in F$, positivity and affine agreement imply

$$0 \le A \le \delta^{-1}P(F)\Phi(b_x)P(F) = \delta^{-1}P(F)\pi(b_x)P(F) \le \frac{M}{\delta}\int_F d_T(\theta(x), \theta(y))^2\, dP(y).$$

If $A \neq 0$, choose a unit vector h such that $\eta = A^{1/2}h \neq 0$. Then $|\eta\rangle\langle\eta| \leq A$ and $\eta \in P(F)H$. Lemma 3.2, applied on $P(F)H$, gives

$$\int_F d_T(\theta(x), \theta(y))^{-2}\, d\mu_\eta(y) \leq M/\delta \qquad (x \in F).$$

The same vector η works for every $x \in F$. Tonelli's theorem therefore gives

$$\iint_{F\times F} d_T(\theta(x), \theta(y))^{-2}\, d\mu_\eta(y) d\mu_\eta(x) \leq (M/\delta)\|\eta\|^2 < \infty,$$

contrary to the energy hypothesis. Thus $A = 0$. Since $A = (E(G)^{1/2}P(F))^*(E(G)^{1/2}P(F))$, it follows that $E(G)P(F) = 0$. □

We will also use the following elementary passage from compact sets to the full spectral measure.

Lemma 3.5. *Let P be a normalized spectral measure and E a normalized POVM on a compact metric space. If $E(B)P(A) = 0$ whenever A, B are disjoint compact sets, then $E = P$.*

Proof. For an open set O, choose compact sets $A_j \uparrow O$. Applying the hypothesis in both orders to A_j and the compact set $X \setminus O$, and taking strong limits, gives

$$E(X \setminus O)P(O) = 0, \qquad E(O)P(X \setminus O) = 0.$$

The two effects sum to I, so $E(O)P(O) = P(O)$ and $E(O)P(X \setminus O) = 0$. Hence $E(O) = P(O)$. The scalar matrix-coefficient measures are regular and agree on open sets; they therefore agree on all Borel sets. □

Proof of Theorem 3.1. Let Φ be an agreeing UCP map. Fix disjoint nonempty compact sets $A, B \subset X$, and put

$$F_m = \left\{ x \in A \cap Z : \min_{y\in B} b_x(y) \geq 1/m \right\}.$$

The minimum over B is a continuous function of the affine coefficients, so F_m is Borel. Strict positivity away from x and compactness of B show that $F_m \uparrow A \cap Z$. Lemma 3.4 gives $E(B)P(F_m) = 0$. Since $P(Z) = I$, strong convergence yields $E(B)P(A) = 0$. Lemma 3.5 now gives $E = P$, and hence $\Phi = \pi$. □

3.2. **A graph consequence.** The criterion applies to represented subsets, independently of the size of the rest of the extreme boundary. The following consequence is useful when a construction begins with a convex graph.

Corollary 3.6. *Let $B \subset \mathbb{R}^D$ be compact and convex, and let g be strictly convex and $C^{1,1}$ on an open convex neighborhood of B. Put*

$$K = \{(a, z) : a \in B,\ g(a) \leq z \leq h\}, \qquad h > \max_B g,$$

and assume $\operatorname{ex} K$ is closed. Let $S \subset B$ be compact and admit covers by $O(\epsilon^{-2})$ sets of diameter at most ϵ. Every representation concentrated on

$$Z = \{(s, g(s)) : s \in S\}$$

has the unique extension property.

Proof. For $x = (s, g(s))$, take the affine barrier

$$b_x(a, z) = z - g(s) - \langle \nabla g(s), a - s\rangle.$$

Strict convexity makes this nonnegative on K, with equality only at x. A Lipschitz bound L for ∇g on B gives

$$b_{(s,g(s))}(t, g(t)) \le \tfrac{L}{2}|s-t|^2 \qquad (s,t \in S).$$

Use $\theta(s, g(s)) = s$. Lemma 3.3 and Theorem 3.1 give the conclusion. □

4. Normal lifts and exposed points in dimension three

Supporting normals provide a canonical metric for the barrier criterion. The useful feature is that they can be added to the boundary points: this places the resulting metric space on the boundary of a parallel convex body. In dimension three, that surface has precisely the critical covering bound of Lemma 3.3.

For $x \in K$, let

$$N_K(x) = \{u \in \mathbb{R}^d : \langle u, y - x\rangle \le 0 \text{ for every } y \in K\}$$

be the outward normal cone. For a unit vector $n_x \in N_K(x)$, put $z_x = x + n_x$. The point z_x lies on $\partial(K + \overline{B}_d)$, and its metric projection onto K is x. Indeed, for $y \in K$,

$$|x + n_x - y|^2 = 1 + |x-y|^2 + 2\langle n_x, x - y\rangle \ge 1,$$

with equality only when $y = x$. For any two such choices,

$$0 \le \langle n_x, x - y\rangle \le \langle n_x - n_y, x - y\rangle \le \tfrac{1}{4}|z_x - z_y|^2. \tag{4.1}$$

The middle inequality uses the supporting property at y; the last follows from $4\langle a, b\rangle \le |a+b|^2$.

Corollary 4.1 (Normal-lift criterion)**.** *Let $\pi : C(X) \to B(H)$ have spectral measure concentrated on a Borel set $Z \subset X$. Suppose a Borel unit normal $n(x)$ exposes x for every $x \in Z$. If every nonzero finite positive Borel measure concentrated on*

$$L = \{x + n(x) : x \in Z\}$$

has infinite inverse-square energy, then π has the unique extension property.

Proof. The map $x \mapsto x + n(x)$ is Borel and injective. Its image is Borel as well: if p_K is the continuous metric projection onto K, then

$$L = \{z \in \partial(K + \overline{B}_d) : p_K(z) \in Z,\ z - p_K(z) = n(p_K(z))\}.$$

Apply Theorem 3.1 with $b_x(y) = \langle n(x), x - y\rangle$ and $\theta(x) = x + n(x)$. The quadratic bound is (4.1), and the energy assumption passes to pushforward measures. □

We need a local version in which the normal need not expose the source point. This is what allows flat faces in the smooth case treated in the next section.

Lemma 4.2 (Separated normal selections)**.** *Let $K \subset \mathbb{R}^3$ be a compact convex body, and suppose Φ and π agree on $A(K)|_X$. Let $F, G \subset X$ be Borel. If there is a Borel unit normal selection $x \mapsto n_x$ on F such that*

$$\langle n_x, x - y\rangle \ge \delta > 0 \qquad (x \in F,\ y \in G),$$

then $E(G)P(F) = 0$.

Proof. Use the affine functions $b_x(y) = \langle n_x, x - y\rangle$ in Lemma 3.4. By (4.1), their values on F are bounded by one quarter of the squared distance between the normal lifts. Every nonzero pushforward measure on these lifts has infinite energy, since the lifts lie on $\partial(K + \overline{B}_3)$ and Lemma 3.3 applies to that surface. □

For exposed points, a separating selection can be made on successively larger compact sets. We record the elementary selection argument to make the measurability explicit.

Lemma 4.3 (Compact-graph selection)**.** *Let F, T be compact metric spaces, and let $\Gamma \subset F \times T$ be compact with every fiber Γ_x nonempty. There is a Borel map $s : F \to T$ with $(x, s(x)) \in \Gamma$ for every $x \in F$.*

Proof. Fix a countable dense sequence in T. At step j, choose the first closed ball of radius 2^{-j} centered in that sequence which meets the current nonempty compact fiber, and intersect the fiber with that ball. These choices are Borel. Indeed, for each fixed finite list of earlier choices, the nonemptiness test is the projection of the original compact graph intersected with finitely many fixed closed balls; this projection is compact. The possible earlier lists form a countable Borel partition, so induction proves the claim. The remaining fibers are nested nonempty compact sets with diameters tending to zero. Their unique intersection point is the limit of the selected centers, and hence depends Borel measurably on x. □

Theorem 4.4 (Exposed extreme boundary)**.** *Let $K \subset \mathbb{R}^3$ be a compact convex body with $X = \operatorname{ex} K$ closed. If every point of X is exposed in K, then $A(K)|_X$ is hyperrigid in $C(X)$.*

Proof. Let π be a unital representation and Φ an agreeing UCP map. Fix disjoint nonempty compact sets $A, B \subset X$. For $m \geq 1$, define

$$\Gamma_m = \left\{ (x, u) \in A \times S^2 : u \in N_K(x), \quad \min_{y \in B} \langle u, x - y \rangle \geq 1/m \right\}, \qquad F_m = \operatorname{pr}_A \Gamma_m.$$

The graph of the normal cone is closed, and the minimum is continuous in (x, u). Thus Γ_m and F_m are compact. The sets F_m increase to A: an exposing normal at x has a strictly positive minimum on the disjoint compact set B. Lemma 4.3 gives a Borel normal selection on F_m with gap at least $1/m$ from B. Lemma 4.2 yields $E(B)P(F_m) = 0$. Taking strong limits gives $E(B)P(A) = 0$, and Lemma 3.5 proves $\Phi = \pi$. □

The selection in this proof depends on the compact target set B. In particular, no continuous choice of exposing normals is required. For a nonexposed extreme point, the compact exhaustion can fail when B meets every supporting face through that point. The next result uses uniqueness of the normal to deal with such faces instead of separating their individual points.

5. Smooth normals and reduction to planar faces

If two smooth boundary points have different normals, the supporting plane at either point strictly separates it from the other. The normal-lift estimate therefore identifies the spectral measure of the normal itself. This is the step that permits a direct-integral argument: the normal algebra must first be shown to lie in the multiplicative domain. Agreement on affine functions alone does not give this. Once the normal is preserved, the fiber representations are supported on planar exposed faces, where the planar hyperrigidity theorem applies.

Theorem 5.1 (Smooth boundary)**.** *Let $K \subset \mathbb{R}^3$ be a compact convex body with C^1 boundary and closed extreme boundary $X = \operatorname{ex} K$. Then $A(K)|_X$ is hyperrigid in $C(X)$.*

We prove a representation-level statement that also applies when only the relevant spectral measures are supported on smooth points. Let

$$S = \{x \in X : N_K(x) \cap S^2 \text{ consists of one point}\}.$$

Thus smoothness here refers to uniqueness of the supporting hyperplane, with no differentiability assumption away from the represented points.

Proposition 5.2. *Let $K \subset \mathbb{R}^3$ be a compact convex body with $X = \operatorname{ex} K$ closed. Let π and ρ be unital representations with spectral measures P and Q, and suppose the UCP map $\Phi(f) = V^* \rho(f) V$ agrees with π on $A(K)|_X$. If $P(S) = I$ and $Q(S) = I$, then $\Phi = \pi$.*

Proof. There are two steps. First the energy estimate forces the dilation isometry to preserve the spectral measure of the normal. This permits a decomposition into representations on individual supporting faces. We then apply the planar theorem on those faces.

The smooth locus. The set-valued map $x \mapsto N_K(x) \cap S^2$ has nonempty compact values and closed graph on ∂K. Compactness of S^2 implies that the diameter of these values is upper semicontinuous. Consequently its singleton locus is Borel, being the intersection of the relatively open sets where that diameter is less than $1/j$. On this locus the unique normal map n is continuous relative to the locus: any subsequential limit of normals is a normal at the limiting point, and uniqueness determines that limit. In particular, $n : S \to S^2$ is Borel.

Separable reduction. It suffices to treat separable representation spaces. If $\Phi \neq \pi$, choose a witness from a countable dense family $\{f_j\} \subset C(X)$ and a vector on which the difference is nonzero. Close its orbit under $\pi(f_j), \Phi(f_j)$ and their adjoints. The resulting separable subspace H_0 reduces both maps and retains the witness. In the given Stinespring space, the closed span of $\rho(C(X))VH_0$ is separable, reduces ρ, and supplies a Stinespring representation for the restricted map. Both spectral concentration assumptions persist under these restrictions. We may therefore assume that H and the Stinespring space are separable.

Choose a common scalar control measure ν for P and Q: all their scalar spectral measures are absolutely continuous with respect to ν. A positive summable weighted sum of the measures of countable dense sets of unit vectors gives such a measure. Normalize it to be a probability measure. It is concentrated on S, and

$$\nu(B) = 0 \quad \Longrightarrow \quad P(B) = Q(B) = 0.$$

Only this implication is needed; no inequality of the form $P(B) \leq \nu(B)I$ is asserted.

Identification of the normal spectral measure. For disjoint compact sets $A, B \subset S^2$, put

$$S_A = S \cap n^{-1}(A), \qquad S_B = S \cap n^{-1}(B).$$

Inner regularity gives increasing compact sets $F_j \subset S_A$ and $G_k \subset S_B$ whose unions have full ν-measure in the respective sets. On $F_j \times G_k$, the function $\langle n(x), x - y\rangle$ is continuous and strictly positive. Indeed, equality would make $n(x)$ a normal at the smooth point y, forcing $n(x) = n(y)$, contrary to the disjoint normal ranges. Its minimum is therefore positive whenever both compact sets are nonempty. Lemma 4.2 gives $E(G_k)P(F_j) = 0$. The sets $S_A \setminus \bigcup_j F_j$ and $S_B \setminus \bigcup_k G_k$ are ν-null, hence null for both spectral measures. Consequently $P(F_j) \to P(S_A)$ and $E(G_k) \to E(S_B)$ strongly. Passing to the limits one at a time gives

$$E(S_B)P(S_A) = 0.$$

Define measures on S^2 by

$$P_n(D) = P(S \cap n^{-1}(D)), \qquad Q_n(D) = Q(S \cap n^{-1}(D)), \qquad E_n(D) = V^*Q_n(D)V.$$

They are normalized because $P(S) = Q(S) = I$. The preceding annihilation and Lemma 3.5 show that $E_n = P_n$. Compression of a projection to a projection gives the intertwining relation

$$Q_n(D)V = VP_n(D) \qquad (D \subset S^2 \text{ Borel}). \tag{5.1}$$

Indeed, $V^*Q_n(D)V = P_n(D)$ and $V^*V = I$ give

$$\|Q_n(D)Vh - VP_n(D)h\|^2 = 0.$$

Thus the entire normal spectral algebra intertwines through V.

Disintegration. The normal map is defined on the possibly nonclosed Borel set S, so we use standard Borel disintegration of measures. Let $\tau = n_*\nu$, with ν regarded as a measure on S. The

regular conditional probability measures ν_ω satisfy

$$\nu(B) = \int_{S^2} \nu_\omega(B)\, d\tau(\omega), \qquad \nu_\omega(S \cap n^{-1}(\{\omega\})) = 1$$

for τ-almost every ω. These are conditional measures on the standard Borel space S, extended by zero to X. To describe the associated representation decomposition explicitly, decompose each separable representation into countably many cyclic representations on $L^2(X, \mu_j)$, with coordinates acting by multiplication. Each μ_j is absolutely continuous with respect to ν. Choose a Borel density $d_j = d\mu_j/d\nu$. For each such cyclic component, the identity

$$\int_X |f|^2\, d\mu_j = \int_{S^2} \left(\int_X |f|^2 d_j\, d\nu_\omega \right) d\tau(\omega)$$

gives its usual disintegration into the spaces $L^2(X, d_j\nu_\omega)$. Applying this to π and ρ gives a common base measure τ and decompositions

$$H = \int_{S^2}^{\oplus} H_\omega\, d\tau(\omega), \qquad L = \int_{S^2}^{\oplus} L_\omega\, d\tau(\omega), \qquad \pi = \int^{\oplus} \pi_\omega, \quad \rho = \int^{\oplus} \rho_\omega.$$

The countably many conditional measures are finite outside one null set. On these fields, the normal projections act as multiplication by $\mathbf{1}_D(\omega)$. Equation (5.1) and the standard characterization of decomposable operators give $V = \int^{\oplus} V_\omega\, d\tau(\omega)$. Uniqueness of decomposable operator fields now turns $V^*V = I$ and the three coordinate compression identities into

$$V_\omega^* V_\omega = I_{H_\omega}, \qquad V_\omega^* \rho_\omega(x_j) V_\omega = \pi_\omega(x_j) \quad (j = 1, 2, 3)$$

outside one common null set. Zero fibers of H can be ignored. This is a decomposition of the representations and their isometry; no pointwise disintegration of an arbitrary UCP map has been assumed.

Planar fibers. For $\omega \in S^2$, let

$$F_\omega = \{x \in K : \langle \omega, x \rangle = h_K(\omega)\}.$$

This exposed face has affine dimension at most two. The conditional spectral measures above are concentrated on $S \cap n^{-1}(\{\omega\}) \subset F_\omega$. Consequently, a continuous function vanishing on $X \cap F_\omega$ acts as zero in both fiber representations. They therefore factor through $C(X \cap F_\omega)$; we keep the same notation for these factor representations. The face identity

$$X \cap F_\omega = \operatorname{ex}(F_\omega)$$

shows that this is the function algebra required by the planar theorem. Indeed, any convex decomposition in K of a point of an exposed face has both endpoints in that face; extremality in the face and in K are therefore equivalent there. Its extreme boundary is compact because X and F_ω are closed. There is no need for the smooth subset of the fiber itself to be closed.

Ambient affine functions restrict onto all affine functions on F_ω. Hence the planar hyperrigidity theorem, including its lower-dimensional cases [Sch1, Theorem 3.8], gives

$$V_\omega^* \rho_\omega(f) V_\omega = \pi_\omega(f) \qquad (f \in C(X \cap F_\omega))$$

for almost every ω. Applying this to the restrictions of a countable dense family in $C(X)$ and integrating proves $\Phi = \pi$. □

Proof of Theorem 5.1. On a C^1 boundary the supporting unit normal is unique, so $S = X$. Every representation and every Stinespring representation are then concentrated on S. Proposition 5.2 applies to every agreeing UCP map, proving hyperrigidity. □

No positive curvature or strict convexity is needed in Theorem 5.1; flat exposed faces are handled by the planar reduction. The concentration hypothesis on the Stinespring representation in Proposition 5.2 is essential to this proof. A nonsmooth dilation point can have several supporting normals, so equality of the normal spectral measures does not follow from the argument above when such points carry spectral mass.

6. An explicit counterexample in affine dimension thirty-three

Theorem 6.1. *There is an explicitly specified compact convex body $K \subset \mathbb{R}^{33}$, with $\operatorname{aff} K = \mathbb{R}^{33}$ and compact extreme boundary $X = \operatorname{ex} K$, for which the affine function system $A(K)|_X$ is not hyperrigid in $C(X)$. More precisely, there are a unital representation $\pi : C(X) \to B(H)$ and a UCP map $\Psi : C(X) \to B(H)$ such that*

$$\Psi|_{A(K)|_X} = \pi|_{A(K)|_X}, \qquad \Psi(Q) \neq \pi(Q), \qquad Q(x,z) = |x|^2.$$

The Hilbert space H is separable and infinite dimensional, the spectral measure of π has no point atoms, and $\dim_{\mathbb{C}} A(K) = 34$.

We use the Walsh dilation of Bilich–Scherer [BS, Section 3], with a different three-point distribution. The operator calculation is included in full. Its telescoping identity determines the height of an encoded dilation point. A uniform scalar estimate places all such points in a truncated epigraph with a strongly convex lower graph. An explicit boundary kernel then gives the required UCP map.

6.1. The Walsh compression and its height identity.

Fix $R > 1$ and $1 < b < R^2$, and put

$$a = b^{-1}, \qquad \mu = \frac{b}{2R^2}(\delta_{-R} + \delta_R) + \left(1 - \frac{b}{R^2}\right)\delta_0.$$

Thus μ is symmetric and has second moment b. In the discrete alphabet $\{-1, 1, -R, 0, R, *\}$, with a separate star symbol, define

$$\begin{aligned} G &= \{-1,1\}^{\mathbb{N}}, \qquad Y_0 = \{-R,0,R\}^{\mathbb{N}}, \\ Y_k &= \{-1,1\}^{k-1} \times \{*\} \times \{-R,0,R\}^{\{k+1,k+2,\dots\}} \quad (k \geq 1), \\ Y &= G \cup \bigcup_{k \geq 0} Y_k. \end{aligned}$$

The numerical coordinate x_n sends the star to zero. Give G Haar probability m. Give Y_k the product probability μ_k with uniform binary prefix, star at k, and μ-distributed tail; on Y_0, use $\mu_0 = \mu^{\otimes \mathbb{N}}$. The space Y is compact: unbounded star positions can accumulate only in G, and every fixed branch is compact. Each coordinate is continuous, and Y_0 is clopen.

Set

$$H = L^2(G, m), \qquad \mathcal{L} = \bigoplus_{k \geq 0} L^2(Y_k, \mu_k).$$

For a finite $F \subset \mathbb{N}$, write $\chi_F = \prod_{n \in F} \varepsilon_n$ and $F_{>k} = F \cap \{k+1, k+2, \dots\}$. For $k \notin F$, the functions

$$\chi_F^{(k)} = b^{-|F_{>k}|/2} \prod_{n \in F} x_n$$

form an orthonormal family in the k-th summand. Define on the Walsh basis

$$V\chi_F = a^{|F|/2}\chi_F^{(0)} + \sqrt{1-a} \sum_{k \in F} a^{|F_{>k}|/2}\chi_{F \setminus \{k\}}^{(k)}. \tag{6.1}$$

Lemma 6.2. *The map $V : H \to \mathcal{L}$ is an isometry. For bounded Borel functions on Y, let $\Phi(f) = V^* M_f V$. Then*

$$\Phi\left(\prod_{n\in E} x_n\right) = M_{\prod_{n\in E}\varepsilon_n} \qquad (E \subset \mathbb{N} \text{ finite}). \tag{6.2}$$

Also $\Phi(\mathbf{1}_{Y_0})\chi_F = a^{|F|}\chi_F$ and $V1 = \mathbf{1}_{Y_0}$.

Proof. On branch k, symmetry and independence make a matrix coefficient of multiplication by $\prod_{n\in E} x_n$ vanish unless the two product labels differ exactly in E and $k \notin E$. In the nonzero case, its value is $b^{|E_{>k}|/2}$: a tail coordinate in the multiplier contributes its second moment divided by $\sqrt{b}$, while common coordinates contribute one after normalization. On branch zero, the same calculation has no star restriction. Consequently, for $C_0 = F \cap F'$,

$$\left\langle M_{\prod_E x_n} V\chi_F, V\chi_{F'}\right\rangle = \mathbf{1}_{\{E=F\triangle F'\}}\left(a^{|C_0|} + (1-a)\sum_{k\in C_0} a^{|(C_0)_{>k}|}\right).$$

The parenthesis is the finite geometric sum $a^r + (1-a)\sum_{j=0}^{r-1} a^j = 1$. Taking $E = \varnothing$ proves isometry on the dense Walsh span; general E proves (6.2). The two final assertions follow by retaining the branch-zero term. □

Let β be an even nonnegative function on the numerical digits, with $\beta(\pm 1) = 0$. Define diagonal contractions

$$A_n\chi_F = a^{|F_{>n}|}\chi_F \qquad (n \geq 0).$$

The parameters $0 < q < 1$ and $c > 0$ will be chosen so that

$$\begin{aligned} \int (r^2 - \beta(r) - 1)\, d\mu(r) &= c(q^2 - 1), \\ a(R^2 - \beta(R) - 1) + (1-a)(-1 - \beta(0)) &= c(q^2 - a). \end{aligned} \tag{6.3}$$

Lemma 6.3. *Under (6.3),*

$$\Phi(x_n^2 - \beta(x_n)) - I = c(q^2 A_n - A_{n-1}) \qquad (n \geq 1). \tag{6.4}$$

Proof. Parity makes the left side diagonal. Fix F and put $r = |F_{>n}|$. If $n \notin F$, the squared weights of the branches before n sum to a^r; their expected multiplier $x_n^2 - \beta(x_n) - 1$ is the first left side of (6.3). Later branches contribute zero, since their n-th coordinate is binary. If $n \in F$, the earlier weights sum to a^{r+1}. Under the normalized x_n^2-weighted measure, the coordinate has magnitude R, so its multiplier is $R^2 - \beta(R) - 1$. The star branch has weight $(1-a)a^r$ and multiplier $-1 - \beta(0)$. The two diagonal coefficients are therefore a^r times the respective left sides of (6.3). These are exactly the coefficients of the right side of (6.4). □

Let D be a positive integer, let $j(n) = 1 + ((n-1) \bmod D)$, and define

$$\begin{aligned} w_n &= q^{n-1} e_{j(n)}, \qquad t(y) = \sum_{n\geq 1} w_n x_n(y), \qquad s = t|_G, \\ U(y) &= \sum_{n\geq 1} q^{2n-2}\beta(x_n(y)), \\ z_Y(y) &= |t(y)|^2 - U(y) + c\mathbf{1}_{Y_0}(y). \end{aligned} \tag{6.5}$$

These series converge uniformly and absolutely. All the displayed functions are continuous, and $U|_G = 0$. The operator calculation already gives

$$\Phi(t_j) = M_{s_j}, \qquad \Phi(z_Y) = M_{|s|^2}. \tag{6.6}$$

Indeed, all mixed terms in the square expansion of $|t|^2$ compress exactly by (6.2). Termwise compression is valid because $\sum_n |w_n| < \infty$. For the diagonal terms, multiplying (6.4) by q^{2n-2} gives the norm-convergent telescope

$$\sum_{n=1}^{N} q^{2n-2}\left[\Phi(x_n^2 - \beta(x_n)) - I\right] = c(q^{2N}A_N - A_0) \longrightarrow -cA_0.$$

This cancels $c\Phi(\mathbf{1}_{Y_0}) = cA_0$, proving (6.6). No orthogonality or cancellation of distinct vectors w_n is used in this calculation.

6.2. **Numerical choices and uniform containment.** The numerical data are collected below; the other quantities retain their definitions above.

Quantity	Value	Quantity	Value
D	32	R	12
b	21/2	q^2	45026/72405
$\beta(0)$	197/200	$\beta(R)$	2417/20
c	14481/24320	Distance coefficient	$1 - 10^{-5}$

In particular $\beta(\pm 1) = 0$ completes the specification of β. The two scalar identities (6.3) are verified exactly: their left sides are respectively

$$-\frac{1441}{6400} = c(q^2 - 1), \qquad \frac{439}{1400} = c(q^2 - a).$$

For the geometric estimate write $\rho = q^D$ and

$$C_\rho = \left\{ \sum_{\ell \geq 0} \rho^\ell \varepsilon_\ell : \varepsilon_\ell \in \{-1, 1\} \right\}.$$

Lemma 6.4 (Uniform digit estimate)**.** *For every sequence $d_\ell \in \{-R, -1, 0, 1, R\}$,*

$$(1 - 10^{-5}) \operatorname{dist}\left(\sum_{\ell \geq 0} \rho^\ell d_\ell, C_\rho \right)^2 \geq \sum_{\ell \geq 0} \rho^{2\ell} \beta(d_\ell). \tag{6.7}$$

Proof. Both sides vanish for an entirely binary sequence. Otherwise let h be its first nonbinary position and put $\delta = \operatorname{dist}(d_h, \{-1, 1\}) \in \{1, R - 1\}$. The largest possible difference between a numerical digit and a binary digit is $R + 1$. Put

$$T = \frac{(R+1)\rho}{1 - \rho}, \qquad E = \frac{\beta(R)\rho^2}{1 - \rho^2}.$$

Against any signed expansion, let $m \leq h$ be the first mismatch. Its leading term dominates the remaining tail and gives a distance at least $\rho^m(|d_m - \varepsilon_m| - T)$. If $m = h$, this is at least $\rho^h(\delta - T)$. If $m < h$, it is at least $\rho^h(R - 1 - T)$, provided $2 - T \geq \rho(R - 1 - T)$. Thus the distance to C_ρ is at least $\rho^h(\delta - T)$. The right side of (6.7) is at most $\rho^{2h}(\beta(d_h) + E)$, since $\beta(R)$ is the largest digit correction.

Here all the required estimates admit rational verification. The integer inequality

$$1990 \cdot 45026^{16} < 72405^{16}$$

gives $\rho < 1/1990$, hence

$$T < \frac{13}{1989} < 1, \qquad E < \frac{2417}{20(1990^2 - 1)}.$$

Also $2-13/1989 > 11/1990$, which implies the earlier-mismatch comparison $2-T > \rho(R-1-T)$. Finally, clearing positive denominators verifies

$$(1-10^{-5})\left(1-\frac{13}{1989}\right)^2 - \frac{197}{200} - \frac{2417}{20(1990^2-1)} > \frac{19}{10000},$$
$$(1-10^{-5})\left(11-\frac{13}{1989}\right)^2 - \frac{2417}{20} - \frac{2417}{20(1990^2-1)} > \frac{49}{10000}.$$

These prove $(1-10^{-5})(\delta-T)^2 > \beta(d_h)+E$ in both cases. The estimates bound the entire infinite tail. □

6.3. **The convex body and its whole extreme boundary.** The represented horizontal set is the Cartesian product

$$C = s(G) = \prod_{j=1}^{D} q^{j-1} C_\rho. \tag{6.8}$$

Define

$$\begin{aligned} g(x) &= |x|^2 - (1-10^{-5})\operatorname{dist}(x,C)^2 \\ &= 10^{-5}|x|^2 + (1-10^{-5})\sup_{u\in C}(2\langle x,u\rangle - |u|^2). \end{aligned} \tag{6.9}$$

The second formula proves strong convexity: it is a positive quadratic plus a supremum of affine functions. With

$$B = [-R-1, R+1]^D, \qquad M = (R+1)^2 D + c + 1,$$

let

$$K = \{(x,z) : x \in B, \quad g(x) \le z \le M\}. \tag{6.10}$$

This set is compact and convex. Since $g(x) \le |x|^2 \le (R+1)^2 D < M$ throughout B, it has nonempty interior in $\mathbb{R}^{D+1}$.

Lemma 6.5. *The entire extreme boundary is the compact set*

$$X = \{(x, g(x)) : x \in B\} \cup \{((R+1)\sigma, M) : \sigma \in \{-1,1\}^D\}. \tag{6.11}$$

Moreover the continuous maps

$$\iota_G(\varepsilon) = (s(\varepsilon), |s(\varepsilon)|^2), \qquad \iota_Y(y) = (t(y), z_Y(y))$$

take their values in X and K, respectively.

Proof. In a convex decomposition of a lower-graph point, equality in the convexity inequality forces the horizontal coordinates to coincide by strict convexity, and then forces the heights to coincide. Every lower-graph point is therefore extreme. An intermediate-height point is a nontrivial vertical convex combination. The top face is a box, whose only extreme points are its vertices. This proves (6.11), including the absence of any other extreme points. Compactness follows from continuity of g and finiteness of the vertex set.

For the containment assertion, $|t_j(y)| \le Rq^{j-1}/(1-\rho) < R+1$. Also $U \ge 0$, so $z_Y \le |t|^2 + c < M$. Apply Lemma 6.4 to each residue class of digits and multiply by $q^{2(j-1)}$. Squared distance to the product (6.8) is the sum of the squared coordinate distances, so

$$(1-10^{-5})\operatorname{dist}(t(y), C)^2 \ge U(y).$$

Consequently $z_Y(y) \ge g(t(y))$, proving $\iota_Y(Y) \subset K$. Finally $g = |\cdot|^2$ on C, so $\iota_G(G) \subset X$. □

6.4. **The boundary kernel and a positive witness.** For $w = (x, z) \in K$, put

$$r(w) = \frac{z - g(x)}{M - g(x)}, \qquad p_\sigma(x) = \prod_{j=1}^{D} \frac{1 + \sigma_j x_j/(R+1)}{2}.$$

The following probability measure on X has barycenter w:

$$\nu_w = (1 - r(w))\delta_{(x,g(x))} + r(w) \sum_{\sigma \in \{-1,1\}^D} p_\sigma(x)\delta_{((R+1)\sigma,M)}. \tag{6.12}$$

Indeed all weights are nonnegative and sum to one, and $\sum_\sigma p_\sigma(x)(R+1)\sigma = x$. The denominator $M - g(x)$ is positive on the whole box. For every $f \in C(X)$, the function $w \mapsto \int f\, d\nu_w$ is continuous.

Define

$$\pi(f) = M_{f \circ \iota_G}, \qquad \widehat{f}(y) = \int_X f\, d\nu_{\iota_Y(y)}, \qquad \Psi(f) = V^* M_{\widehat{f}} V. \tag{6.13}$$

The first map is a unital representation. Pointwise integration against probability measures preserves positivity at every matrix level; multiplication and isometric compression therefore make Ψ UCP. This formula uses the explicit finite kernel (6.12), with no choice of representing measures.

It also gives a representation dilation directly. Index the terms of (6.12) by $J = \{0\} \cup \{-1, 1\}^D$, writing their locations and weights at $\iota_Y(y)$ as $F_\ell(y) \in X$ and $\omega_\ell(y) \geq 0$. On $\widetilde{\mathcal{L}} = \bigoplus_{k \geq 0} \bigoplus_{\ell \in J} L^2(Y_k, \mu_k)$, let $\rho_X(f)$ multiply component (k, ℓ) by $f(F_\ell(y))$. The map

$$(W\xi)_{k,\ell}(y) = \sqrt{\omega_\ell(y)}\, \xi_k(y)$$

is an isometry, and $\Psi(f) = (WV)^* \rho_X(f)(WV)$.

Completion of the proof of Theorem 6.1. For an affine function, its integral against ν_w is its value at w. Thus (6.6) gives

$$\Psi(x_j|_X) = M_{s_j} = \pi(x_j|_X), \qquad \Psi(z|_X) = M_{|s|^2} = \pi(z|_X).$$

Together with constants, these coordinates span $A(K)|_X$. They are linearly independent because K has full affine dimension $D + 1 = 33$, and restriction to X is injective because $K = \operatorname{conv} X$. Hence the function system has complex dimension $D + 2 = 34$.

For $Q(x, z) = |x|^2$, scalar convexity and the barycenter property give $\widehat{Q}(y) \geq |t(y)|^2$. The constant Walsh vector has norm one and $V1 = \mathbf{1}_{Y_0}$. Independence and zero means remove all mixed terms from the following scalar integrals:

$$\langle \Psi(Q)1, 1\rangle \geq \int_{Y_0} |t|^2\, d\mu_0 = \frac{b}{1 - q^2},$$
$$\langle \pi(Q)1, 1\rangle = \int_G |s|^2\, dm = \frac{1}{1 - q^2}.$$

Consequently

$$\langle [\Psi(Q) - \pi(Q)]1, 1\rangle \geq \frac{b - 1}{1 - q^2} > 0.$$

The representation therefore fails the unique extension property.

Finally $q^D = \rho < 1/2$ makes each signed expansion in C_ρ injective: the first differing digit exceeds the entire remaining difference tail. The product encoding s, and hence ι_G, is injective. A point inverse image under ι_G is therefore empty or a singleton; Haar measure on G assigns it measure zero. This proves that the spectral measure of π has no point atoms. The countably infinite Walsh basis proves that H is separable and infinite dimensional. □

7. An obstruction for stationary geometric Walsh encodings

The construction above separates the operator identities from the geometric containment problem. We now show that its dimension cannot be reduced to four by changing only the fixed three-point distribution, the geometric decay rate, or the directions of the encoding vectors. The first obstruction concerns the quadratic distance envelope. The next section extends it, in affine dimension four, to every finite strictly convex lower graph.

7.1. **A signing estimate.**

Lemma 7.1 (Signed series in finite dimension)**.** *Let $w_n \in \mathbb{R}^D$ satisfy $\sum_n |w_n| < \infty$, and let L_D be the sum of the D largest numbers among $\{|w_n|^2 : n \geq 1\}$, padding with zeros if necessary. There are signs $\varepsilon_n \in \{-1, 1\}$ such that*

$$\left| \sum_{n \geq 1} \varepsilon_n w_n \right|^2 \leq L_D.$$

Proof. For a finite sequence $w_1, \ldots, w_N$, the polytope

$$\left\{ a \in [-1,1]^N : \sum_{n=1}^{N} a_n w_n = 0 \right\}$$

is nonempty. At an extreme point a, at most D coordinates lie strictly between -1 and 1: otherwise the corresponding vectors are linearly dependent and admit a nonzero feasible perturbation in both directions. Independently round each of these coordinates to a sign with mean a_n, leaving the other coordinates fixed. Since the centered rounding errors are independent,

$$\mathbb{E} \left| \sum_{n=1}^{N} \varepsilon_n w_n \right|^2 = \sum_{|a_n|<1} (1 - a_n^2)|w_n|^2 \leq L_D.$$

Some realization therefore has the required bound. Apply this argument to finite prefixes and extend their signings arbitrarily. The remaining tail has norm at most $\sum_{n>N} |w_n| \to 0$. Compactness of the sign space and uniform convergence of the signed series give the assertion for the infinite sequence. □

7.2. **Eliminating all fixed three-point parameters.**

In this section, γ denotes the squared geometric ratio; thus $\gamma = q^2$ in the construction above. The numerical tail law is supported on $\{0, \pm R\}$, with variance b and masses $1 - b/R^2, b/(2R^2), b/(2R^2)$. As before, G is the binary source, Y_0 has only tail coordinates, and Y_k has a binary prefix, a zero-valued star at k, and tail coordinates thereafter.

Theorem 7.2 (No stationary geometric distance encoding in horizontal dimension at most four)**.** *Let*

$$R > 1, \qquad 1 < b < R^2, \qquad c > 0, \qquad 0 < \gamma < 1, \qquad u, v \geq 0.$$

Suppose the exact height equations hold:

$$\begin{aligned} \frac{b}{R^2}(R^2 - 1 - v) + \left(1 - \frac{b}{R^2}\right)(-1 - u) &= c(\gamma - 1), \\ \frac{1}{b}(R^2 - 1 - v) + \left(1 - \frac{1}{b}\right)(-1 - u) &= c(\gamma - 1/b). \end{aligned} \tag{7.1}$$

Let $w_n \in \mathbb{R}^D$ have arbitrary directions and satisfy $|w_n|^2 = \gamma^{n-1}$. Define

$$t(y) = \sum_{n \geq 1} w_n x_n(y), \qquad C = t(G), \qquad U(y) = \sum_{n \geq 1} \gamma^{n-1} \beta(x_n(y)),$$

where $\beta(\pm 1) = 0$, $\beta(0) = u$, and $\beta(\pm R) = v$. If $D \le 4$, there is no $0 < \lambda \le 1$ for which

$$\lambda \operatorname{dist}(t(y), C)^2 - U(y) + c\mathbf{1}_{Y_0}(y) \ge 0 \qquad (y \in Y). \tag{7.2}$$

Proof. Two consequences of Lemma 7.1 suffice. At a star in the first position followed by zero digits, the encoded point is zero and $U = u/(1-\gamma)$. Signing the source series bounds its distance to C. On Y_0, take digits $R\varepsilon_n$ and compare with the source point having digits ε_n; the difference is $(R-1)\sum_n \varepsilon_n w_n$. Thus (7.2) implies, after discarding $\lambda \le 1$,

$$u \le 1 - \gamma^D, \qquad v - c(1-\gamma) \le (R-1)^2(1-\gamma^D). \tag{7.3}$$

For $D \le 4$ these inequalities remain necessary with D replaced by four.

Put $x = \gamma^4$, $h = 1 + u$, and $\Delta = R^2 + u - v$. The height equations give

$$c = \frac{\Delta(R^2 - b^2)}{R^2(b-1)}, \qquad c(1-\gamma) = h - \frac{b}{R^2}\Delta. \tag{7.4}$$

Using the second equality in the second inequality of (7.3) yields

$$\left(1 - \frac{b}{R^2}\right)\Delta \ge 2(R-1) + (R-1)^2 x > 0. \tag{7.5}$$

Hence $\Delta > 0$, and the first equality of (7.4), together with $c > 0$, forces $b < R$. In particular all subsequent denominators are positive. Eliminating c also gives

$$\frac{h}{\Delta} = \frac{R^2 - b - \gamma(R^2 - b^2)}{R^2(b-1)} > 0.$$

Multiply (7.5) by h/Δ and use $h \le 2 - x$, from the first inequality in (7.3). After cancellation,

$$(R-1)(2 + (R-1)x)\left(1 - \gamma\frac{R^2 - b^2}{R^2 - b}\right) \le (b-1)(2-x). \tag{7.6}$$

Set $r = R - 1 > 0$, $\alpha = (b-1)/(R-1) \in (0,1)$, and $A = 1 - \alpha \in (0,1)$. Clearing the positive denominators in (7.6) says that $p(r) \ge 0$, where

$$p(r) = \gamma(1-\alpha)(2 + (1+\alpha)r)(2 + rx) - (r + 2 - \alpha)(2(1-\alpha) + (r+\alpha)x).$$

Its quadratic, linear, and constant coefficients are respectively

$$\begin{aligned} &x(\gamma(1-\alpha^2) - 1), \\ &2\big[-\gamma A^2 + (2\gamma - 1 + \gamma x)A - x\big], \\ &-(2-x)A^2 + 2(2\gamma - 1)A - x. \end{aligned} \tag{7.7}$$

We prove that all three are strictly negative. The first is immediate. For the second put $N = 2\gamma - 1 + \gamma^5$. If $N \le 0$, negativity is immediate; if $N > 0$, half the coefficient is at most $N^2/(4\gamma) - \gamma^4$. We claim that

$$2\gamma - 1 + \gamma^5 < 2\gamma^{5/2} \qquad (0 < \gamma < 1). \tag{7.8}$$

Indeed, setting $t = \sqrt{\gamma}$, the difference factors as

$$t^{10} - 2t^5 + 2t^2 - 1 = (t-1)Q(t),$$

where

$$Q(t) = t^9 + t^8 + t^7 + t^6 + t^5 - t^4 - t^3 - t^2 + t + 1 > 0 \qquad (0 < t < 1).$$

For $t \le 2/3$, this follows from $t^2 + t^3 + t^4 \le 76/81 < 1$. For $t \ge 2/3$, use

$$Q(t) = (1-t^2)(1+t) + t^4(t + t^2 + t^3 + t^4 + t^5 - 1),$$

whose first term is positive and whose second is nonnegative because $t + t^2 \geq 10/9$. This proves (7.8) and negativity of the linear coefficient.

For the constant coefficient in (7.7), $\gamma \leq 1/2$ is immediate. If $\gamma > 1/2$, its maximum over real A is

$$\frac{(2\gamma - 1)^2}{2 - \gamma^4} - \gamma^4 < 0.$$

To verify the strict inequality, factor its numerator as

$$(2\gamma - 1)^2 - 2\gamma^4 + \gamma^8 = (\gamma - 1)(\gamma^7 + \gamma^6 + \gamma^5 + \gamma^4 - \gamma^3 - \gamma^2 + 3\gamma - 1).$$

The parenthesis is positive, since $3\gamma - 1 - \gamma^2 - \gamma^3 = (1 - \gamma)(\gamma^2 + 2\gamma - 1) \geq 0$ on this interval and the remaining powers are positive. All coefficients of p are therefore negative, so $p(r) < 0$ for $r > 0$, a contradiction. □

Theorem 7.2 permits arbitrary, nonperiodic directions and every fixed symmetric three-point distribution with nonnegative potential satisfying the stated height equations. It excludes the distance-envelope realization in affine dimensions four and five, but is not a universal lower bound for commutative counterexamples. The remaining possibility of changing the lower graph requires an operator argument.

8. Strictly convex graphs in affine dimension four

The quadratic envelope is forced by an injective affine defect when the horizontal dimension is at most three. The proof requires no smoothness of the lower graph. Away from its quadratic tangencies, supporting affine functions admit a metric taking values on a positive distance level. Such levels have the critical covering bound needed by Theorem 3.1.

8.1. **Distance levels and supporting affine functions.**

Lemma 8.1 (Covering a positive distance level)**.** *Let $C \subset \mathbb{R}^D$ be nonempty and compact, and fix $\varepsilon > 0$. The compact set*

$$S_\varepsilon = \{z : \operatorname{dist}(z, C) = \varepsilon\}$$

has covers by $O(r^{-(D-1)})$ sets of diameter at most r as $r \downarrow 0$.

Proof. For $z \in S_\varepsilon$, choose a nearest point $c \in C$ and put $v = (z - c)/\varepsilon$. If $z' \in S_\varepsilon$ and $d = z' - z$, then $|z' - c| \geq \varepsilon$. Interchanging the two points gives, for any nearest-point unit vector v' at z',

$$v \cdot d \geq -\frac{|d|^2}{2\varepsilon}, \qquad v' \cdot d \leq \frac{|d|^2}{2\varepsilon}. \tag{8.1}$$

Cover S_ε by finitely many closed spatial sets of diameter at most $\varepsilon/4$, and the unit sphere by finitely many closed balls of radius $1/8$ with unit centers e_j. Partition into overlapping patches by requiring membership in one spatial set and the existence of a nearest-point vector within $1/8$ of one e_j. These patches are compact, since they are projections of compact sets of nearest-point pairs.

For two points in the same patch, choose witnessing vectors in (8.1). Then

$$|e_j \cdot d| \leq \frac{1}{8}|d| + \frac{|d|^2}{2\varepsilon} \leq \frac{1}{4}|d|.$$

Orthogonal projection onto $e_j^\perp$ is therefore injective on the patch and has a Lipschitz inverse. A grid in its bounded projected subset of $\mathbb{R}^{D-1}$ gives the stated covering bound. Sum over the finitely many patches. For $D = 1$, each patch contains at most one point. No continuity or uniqueness of nearest-point vectors is needed. □

For a finite convex function g on an open convex set Ω, we write

$$\partial g(s) = \{p : \ g(t) \geq g(s) + \langle p, t-s\rangle \text{ for all } t \in \Omega\}$$

for its convex subdifferential. The next lemma uses subgradients that differ from the gradient $2s$ of the quadratic source graph.

Lemma 8.2 (A metric for subgradient barriers)**.** *Let g be finite and convex on an open convex set $\Omega \subset \mathbb{R}^D$, and let $C \subset \Omega$ be compact with $g(s) = |s|^2$ on C. Suppose $F \subset C$ is Borel and $p_s \in \partial g(s)$ is a Borel selection satisfying $0 < \delta \leq |p_s - 2s| \leq M < \infty$ on F. Set*

$$v_s = \frac{p_s - 2s}{|p_s - 2s|}, \qquad \varepsilon = \delta/16, \qquad \theta_s = s + \varepsilon v_s.$$

Then $\theta : F \to S_\varepsilon$ is Borel, with the distance level taken relative to the whole set C, and for a finite constant L depending only on δ, M,

$$0 \leq |t|^2 - |s|^2 - p_s \cdot (t-s) \leq L|\theta_t - \theta_s|^2 \qquad (s,t \in F). \tag{8.2}$$

Proof. The subgradient inequality and $g = |\cdot|^2$ on C give

$$(p_s - 2s)\cdot(t-s) \leq |t-s|^2 \qquad (s \in F,\ t \in C).$$

Consequently

$$|\theta_s - t|^2 \geq \varepsilon^2 + (1 - 2\varepsilon/\delta)|t-s|^2.$$

Thus s is the unique nearest point in C to θ_s, and $\theta_s \in S_\varepsilon$.

For $s,t \in F$ write $d = t - s$ and $w = v_t - v_s$. Applying the subgradient inequality in both orders gives

$$v_s \cdot d \leq |d|^2/\delta, \qquad v_t \cdot d \geq -|d|^2/\delta,$$

so $w \cdot d \geq -2|d|^2/\delta$ and

$$|\theta_t - \theta_s|^2 = |d + \varepsilon w|^2 \geq \tfrac{3}{4}|d|^2 + \varepsilon^2|w|^2. \tag{8.3}$$

The barrier in (8.2) is nonnegative by convexity. Writing $\alpha_s = |p_s - 2s| \leq M$, its upper bound is

$$|d|^2 - \alpha_s v_s \cdot d \leq (1 + M/\delta)|d|^2 + M|w|\,|d|.$$

Young's inequality and (8.3) bound this by $L|\theta_t - \theta_s|^2$ for a finite L. □

8.2. **An injective defect forces the quadratic envelope.**

Theorem 8.3 (An injective defect forces the quadratic envelope)**.** *Let $1 \leq D \leq 3$, and let g be finite and strictly convex on an open convex set $\Omega \subset \mathbb{R}^D$. Let $K \subset \Omega \times \mathbb{R}$ be compact and convex, with $X = \operatorname{ex} K$ compact and $K \subset \{(t,z) : z \geq g(t)\}$. Suppose $C \subset \Omega$ is nonempty and compact, and*

$$g(s) = |s|^2, \qquad x_s = (s, |s|^2) \in K \qquad (s \in C).$$

The graph points form a set $E = \{x_s : s \in C\} \subset X$. Let $\pi : C(X) \to B(H)$ be a unital representation concentrated on E and having full spectral support there. Suppose a UCP map $\Psi : C(X) \to B(H)$ agrees with π on affine functions, and for some real affine h the operator

$$\mathcal{D}_h = \Psi(h^2) - \pi(h)^2$$

is injective. Then π is concentrated on points where $\partial g(s) = \{2s\}$, equivalently where g is differentiable with $\nabla g(s) = 2s$. Moreover,

$$2s \in \partial g(s) \quad (s \in C), \qquad g(t) \geq |t|^2 - \operatorname{dist}(t,C)^2 \quad (t \in \Omega). \tag{8.4}$$

Proof. Subdifferentials of a finite convex function on an open convex domain are nonempty and compact, locally bounded, and have closed graph. Local boundedness follows by evaluating the supporting inequality a fixed small distance in every direction inside a larger compact ball. Hence the subgradient graph over C is compact, and $|p-2s|\le M$ there for some finite M.

For $m\ge 1$ let

$$\Gamma_m=\{(s,p): s\in C,\ p\in\partial g(s),\ |p-2s|\ge 1/m\},\qquad F_m=\mathrm{pr}_C\,\Gamma_m.$$

These sets are compact. The increasing sets F_m exhaust $\{s\in C:\partial g(s)\neq\{2s\}\}$. On each nonempty F_m, Lemma 4.3 gives a Borel selector with $1/m\le|p_s-2s|\le M$. For the graph $E_m=\{x_s:s\in F_m\}$ define

$$b_{x_s}(t,z)=z-|s|^2-p_s\cdot(t-s).$$

The coefficients are Borel, and b_{x_s} is nonnegative on K and vanishes there only at x_s. Indeed strict convexity makes the subgradient inequality strict at every $t\neq s$; at $t=s$, equality requires $z=g(s)$. This also shows that all graph points under consideration are exposed extreme points of K.

Lemma 8.2, with $\delta=1/m$, bounds these barriers on E_m by a constant times squared distance after a Borel map into $S_{1/(16m)}$. By Lemma 8.1, this compact level has $O(r^{-(D-1)})$ covers and hence, since $D\le 3$, $O(r^{-2})$ covers. Lemma 3.3 gives infinite inverse-square energy for every nonzero finite measure pushed onto it. Theorem 3.1 therefore gives UEP for every representation concentrated on E_m.

Write P for the spectral measure of π and $P_m=P(E_m)$. Restrict π and compress Ψ to each nonzero P_mH. These maps still agree on the affine functions, so the preceding UEP assertion makes them equal. Since P_m commutes with $\pi(h)$,

$$P_m\mathcal{D}_hP_m=0.$$

The operator $\mathcal{D}_h$ is positive by the Schwarz inequality. Thus $\mathcal{D}_h^{1/2}P_m=0$, and injectivity forces $P_m=0$. The Borel set

$$E_{\mathrm{bad}}=\bigcup_m E_m=\{x_s:s\in C,\ \partial g(s)\neq\{2s\}\}$$

has spectral projection zero. A finite convex function is differentiable precisely where its subdifferential is a singleton, which proves the concentration assertion.

Full spectral support makes the remaining source points dense in C. The closed subgradient graph then gives $2s\in\partial g(s)$ for every $s\in C$. Finally,

$$g(t)\ge\sup_{s\in C}(g(s)+2s\cdot(t-s))=|t|^2-\mathrm{dist}(t,C)^2,$$

proving (8.4). □

Corollary 8.4 (UEP away from quadratic tangencies)**.** *Under the geometric hypotheses of Theorem 8.3, every representation concentrated on $E_{\mathrm{bad}}=\{x_s:s\in C,\ \partial g(s)\neq\{2s\}\}$ has UEP. More generally, for any representation π concentrated on E and any affine-agreeing UCP map Ψ,*

$$(\Psi(h^2)-\pi(h)^2)P(E_{\mathrm{bad}})=0$$

for every real affine h.

Proof. Use the increasing compact sets E_m in the preceding proof, each of which supports only representations with UEP. If $P(E_{\mathrm{bad}})=I$, then $P(E_m)\uparrow I$ and UEP on the compressed spaces gives

$$P(E_m)\Psi(f)P(E_m)=P(E_m)\pi(f)P(E_m)\qquad(f\in C(X)).$$

Taking strong limits proves $\Psi = \pi$. Without that concentration assumption, compressed equality still gives $P(E_m)\mathcal{D}_h P(E_m) = 0$. Positivity yields $\mathcal{D}_h P(E_m) = 0$, and the strong limit gives the result. □

8.3. **Application to the Walsh construction.**

Lemma 8.5 (Injective horizontal defects)**.** *For the stationary three-point Walsh compression, assume $1 < b < R^2$ and write $\Phi(f) = V^* M_f V$. Each operator $D_n = \Phi(x_n^2) - I$ is positive and injective. If $t_j = \sum_n (w_n)_j x_n$ is a nonzero linear coordinate with $\sum_n R|w_n| < \infty$ and $s_j = t_j|_G$, then*

$$\Phi(t_j^2) - M_{s_j}^2 = \sum_n (w_n)_j^2 D_n \tag{8.5}$$

is also positive and injective. A positive barycentric-kernel realization with horizontal coordinate t_j has an actual coordinate defect at least as large as this operator.

Proof. Put $a = 1/b$. On the Walsh vector χ_F, let $r = |F_{>n}|$. If $n \notin F$, the total squared weight of branches before n is a^r, and their tail second moment is b; later branches have binary value squared equal to one. If $n \in F$, the total weight before n is a^{r+1}, their squarefree-product-weighted second moment is R^2, and branch n has weight $(1-a)a^r$ and value zero. Consequently

$$D_n \chi_F = a^r \begin{cases} b - 1, & n \notin F, \\ R^2/b - 1, & n \in F. \end{cases} \chi_F.$$

Every diagonal entry is strictly positive, so D_n is injective; no uniform positive lower bound is asserted.

Uniform absolute convergence permits expansion of t_j^2. All mixed terms compress exactly by (6.2), leaving the norm-convergent positive sum (8.5). One nonzero coefficient bounds this sum below by a positive scalar multiple of an injective D_n. Finally, scalar Jensen for a probability measure with barycenter $(t(y), z(y))$ gives $\int \tau_j^2 \, d\nu_y \geq t_j(y)^2$. Applying the positive compression shows $\Psi(\tau_j^2) \geq \Phi(t_j^2)$, while affine agreement gives $\pi(\tau_j) = M_{s_j}$. This proves the last assertion. □

Corollary 8.6 (No strictly convex graph in affine dimension four)**.** *Take the stationary geometric data of Theorem 7.2. Suppose a compact convex body K with compact extreme boundary contains every encoded point*

$$(t(y), z_Y(y)), \qquad z_Y(y) = |t(y)|^2 - U(y) + c\mathbf{1}_{Y_0}(y), \qquad y \in Y,$$

and lies in the epigraph of a finite strictly convex function g on an open convex horizontal domain containing $t(Y)$. Assume $g(s) = |s|^2$ on $C = t(G)$, the signed graph is represented on $L^2(G, m)$, and a positive barycentric kernel realizes an affine-agreeing UCP map. Then $D \geq 4$. In particular this mechanism cannot give an affine-dimension-four counterexample, with or without differentiability of g.

Proof. The continuous surjection $t : G \to C$ sends Haar measure to a measure of full support: the preimage of a nonempty relatively open set is nonempty and open, and therefore has positive Haar measure. This does not require injectivity of the encoding. Some horizontal coordinate has a nonzero coefficient, and Lemma 8.5 gives an injective defect for the actual barycentric map.

If $D \leq 3$, Theorem 8.3 gives $g(t) \geq |t|^2 - \operatorname{dist}(t, C)^2$. Containment of the encoded points therefore implies

$$\operatorname{dist}(t(y), C)^2 - U(y) + c\mathbf{1}_{Y_0}(y) \geq 0 \qquad (y \in Y),$$

contradicting Theorem 7.2 with $\lambda = 1$. □

Remark 8.7. Theorem 8.3 itself does not assume a Walsh model, stationarity, or geometric vector norms. It uses one injective affine defect and full spectral support on the quadratic graph. Its dimension restriction is $D - 1 \leq 2$, the covering exponent of the auxiliary distance level. Accordingly, the strictly convex graph obstruction applies in affine dimension four; the stronger affine-dimension-five exclusion in Theorem 7.2 applies to the distance-envelope certificate. These are construction-family obstructions, not universal dimension lower bounds. They leave open constructions using a different compression, nongeometric vector norms, other height identities, or geometry outside the strictly convex graph setting.

9. The remaining low-dimensional questions

The main results leave both parts of the proposed dimension-four threshold unresolved. There is no construction here in affine dimension four, and the positive three-dimensional theorems do not cover every compact convex body with closed extreme boundary.

In dimension three, one unresolved interaction in the smooth-source approach is between smooth source points and nonsmooth dilation points. Different source normals may have supporting faces containing the same dilation point. The proof of Proposition 5.2 then gives no intertwining over the normal space. Distinct supporting faces in $\mathbb{R}^3$ intersect in at most a line and have at most two common extreme points, but pointwise affine interpolation on these intersections is not yet an operator estimate. In particular, a bounded array of interpolation coefficients need not define a bounded Schur multiplier uniformly in the number of spectral blocks. If H_N is a symmetric Hadamard matrix and C_N agrees with it off the diagonal and vanishes on the diagonal, then

$$\|H_N/\sqrt{N}\| = 1, \qquad C_N \circ (H_N/\sqrt{N}) = (J_N - I_N)/\sqrt{N},$$

whose norm is $(N-1)/\sqrt{N}$. Any argument along this route must use the actual incidence constraints of the supporting faces, not only entrywise coefficient bounds.

For dimension four, the signing and proximal arguments rule out the most direct ways of modifying the stationary geometric Walsh construction. The expectation obstruction in Appendix C rules out the quadratic-distance containment certificate for every exact height correction for the first correlated distribution. These results identify failures of specific mechanisms; they provide no evidence that all four-dimensional bodies must be hyperrigid. A successful construction must simultaneously preserve exact compression identities, place the represented spectrum in the extreme boundary, and retain a nonzero coordinate defect. None of these requirements follows from the others.

Preparation. Separate automated review tasks rederived the compression identities, checked the geometric and measure-theoretic arguments, and tested proposed constructions against the stated hypotheses. Exact rational arithmetic verifies the numerical certificate in Section 6. The bibliography identifies the source versions used; the limited literature comparison does not establish priority for every subsidiary lemma. The presentation separates complete arguments from the open questions above.

Appendix A. Normal-lift capacity is not a necessary condition

The normal lift gives a convenient metric for the supporting-gap criterion, but it need not give the strongest one. The next example separates the two tests. Its represented source has the unique extension property even though its normal lift supports a measure of finite inverse-square energy.

Theorem A.1. *There are a compact convex body $K \subset \mathbb{R}^4$ with closed extreme boundary and a compact set $Z \subset \operatorname{ex} K$ such that:*

(i) *every point of Z is exposed and has a unique unit outward normal;*
(ii) *the normal lift of Z supports a probability measure of finite inverse-square energy;*
(iii) *every representation concentrated on Z has UEP relative to $A(K)|_{\operatorname{ex} K}$.*

Proof. We construct a thin Cantor set whose normal field records two additional independent collections of digits. For signs $\omega_n \in \{-1,1\}$, $n \ge 0$, put

$$s(\omega) = \left(\sum_{n\ge 0} 4^{-n}\omega_n, 10^{-3}\sum_{n\ge 0} 64^{-n}\omega_n, 10^{-3}\sum_{n\ge 0} 128^{-n}\omega_n \right), \qquad C = s(\{-1,1\}^{\mathbb{N}_0}).$$

The first coordinate is an injective Cantor expansion, so s is a homeomorphism onto C. Its first three vector coefficients are linearly independent by the Vandermonde determinant; hence $\operatorname{aff} C = \mathbb{R}^3$. Also $C \subset (-2,2)^3$ and $|s| < 2$.

If two sequences first differ at index n, their first-coordinate difference is at least $(4/3)4^{-n}$, whereas their second- and third-coordinate differences are at most $2\cdot 10^{-3}64^{-n}/(1-1/64)$ and $2\cdot 10^{-3}128^{-n}/(1-1/128)$, respectively. In particular,

$$|s_j - t_j| \le \frac{8}{7000}|s_1 - t_1|^2 \quad (j=2,3), \qquad |s-t|^2 \le 2|s_1-t_1|^2. \tag{A.1}$$

Define continuous functions $a, b : C \to [1,2]$ by

$$a(s(\omega)) = 1 + \sum_{k\ge 0} 2^{-(k+1)}\frac{1+\omega_{3k+1}}{2}, \qquad b(s(\omega)) = 1 + \sum_{k\ge 0} 2^{-(k+1)}\frac{1+\omega_{3k+2}}{2}.$$

Set $p(s) = 2s + (0, a(s), b(s))$. The preceding estimates imply

$$\big|\langle p(s) - 2s, t - s\rangle\big| \le \frac{32}{7000}|t-s|^2 < \frac{1}{2}|t-s|^2 \qquad (s,t \in C). \tag{A.2}$$

Thus the data $g(s) = |s|^2$, $\nabla g(s) = p(s)$ admit the strongly convex extension

$$g(x) = \max_{s\in C}\left\{ |s|^2 + \langle p(s), x - s\rangle + \tfrac{1}{2}|x-s|^2 \right\}.$$

Indeed, subtracting $|x|^2/2$ leaves a supremum of affine functions. For $x \in C$, (A.2) shows that the unique maximizer is $s = x$, and $g(x) = |x|^2$.

For clarity, this uniqueness also proves differentiability at every $x \in C$. If s_h maximizes at $x+h$, compactness and continuity force $s_h \to x$. Let Q_s denote the quadratic inside the maximum. Then

$$Q_x(x+h) - Q_x(x) \le g(x+h) - g(x) \le Q_{s_h}(x+h) - Q_{s_h}(x).$$

Both bounds are $\langle p(x), h\rangle + o(|h|)$.

Choose $M > \max_{[-2,2]^3} g$ and let

$$K = \{(x,z) : x \in [-2,2]^3,\ g(x) \le z \le M\}, \qquad Z = \{P_s = (s, |s|^2) : s \in C\}.$$

Strict convexity shows that the full extreme boundary consists of the lower graph and the eight vertices of the top face, exactly as in Lemma 6.5. It is compact, and K has nonempty interior. At P_s, the supporting plane is strict away from P_s, and the unique outward unit normal is

$$n(s) = \frac{(p(s), -1)}{\sqrt{1 + |p(s)|^2}}.$$

We next show that the lift $P_s + n(s)$ carries finite energy. The paired lift $(P_s, n(s))$ and the sum lift $P_s + n(s)$ are bi-Lipschitz equivalent: the supporting-normal inequalities give

$$|P_s - P_t|^2 + |n(s) - n(t)|^2 \le |P_s + n(s) - P_t - n(t)|^2 \le 2\big(|P_s - P_t|^2 + |n(s) - n(t)|^2\big).$$

By (A.1), the paired lift is in turn bi-Lipschitz equivalent to the coordinates $(s_1, a(s), b(s))$. In one direction all its components are Lipschitz functions of these three coordinates. In the other, recover s from the position and then recover $p(s)$ from the normal by division by its last coordinate, which is bounded away from zero.

Under product measure on the signs, a and b are independent uniform variables on $[1, 2]$. Conditional on the digits defining them, the remaining contribution to s_1 is a translate of

$$\sum_{k\geq 0} 64^{-k}\omega_{3k}.$$

Its probability in an interval of length $2r$ is bounded by a fixed multiple of $r^{1/6}$. To verify this, choose N with $64^{-N} \leq r < 64^{-(N-1)}$. An interval of that length meets at most two level-N cylinders, each of mass 2^{-N}; the separation follows by comparing the first differing digit with the geometric tail. Conditioning and integrating over a, b therefore give a ball bound of the form $Cr^{2+1/6}$ for (s_1, a, b), uniformly in the center. The same bound holds on the normal lift by bi-Lipschitz equivalence. For its pushed-forward probability measure ν, it follows that

$$\iint |u - v|^{-2}\, d\nu(u)d\nu(v) \leq 1 + 2C \int_0^1 r^{-3}r^{13/6}\, dr < \infty.$$

Finally, consider the actual supporting affine functions

$$b_{P_s}(x, z) = z - |s|^2 - \langle p(s), x - s\rangle.$$

They expose P_s, since $b_{P_s}(x, z) \geq z - g(x) + |x - s|^2/2$ on K. On the represented source, (A.1)–(A.2) give

$$0 \leq b_{P_s}(P_t) \leq 3|s_1 - t_1|^2.$$

The first coordinate lies in the compact interval $[-4/3, 4/3]$, which supports no nonzero measure of finite inverse-square energy. Theorem 3.1 proves the unique extension property on Z. □

Appendix B. Varying distributions and uniformly positive defects

The local defect estimate in Lemma 2.3 imposes a restriction independent of the choice of convex graph. Here is its consequence for varying three-point distributions.

At coordinate n, let the symmetric distribution on $\{0, \pm R_n\}$ have second moment b_n, where $1 < b_n < R_n^2$. Put $a_n = b_n^{-1}$. Use the branch spaces of the Walsh construction, with these varying tail distributions. Replace the formula for V by

$$\begin{aligned} V\chi_F = &\left(\prod_{n\in F} \sqrt{a_n}\right) \chi_F^{(0)} \\ &+ \sum_{k\in F} \sqrt{1 - a_k} \left(\prod_{n\in F_{>k}} \sqrt{a_n}\right) \chi_{F\setminus\{k\}}^{(k)}, \end{aligned}$$

where the normalized branch monomial is $\chi_F^{(k)} = (\prod_{n\in F_{>k}} b_n^{-1/2}) \prod_{n\in F} x_n$. The proof of Lemma 6.2 remains valid because

$$\prod_{n\in F} a_n + \sum_{k\in F}(1 - a_k) \prod_{n\in F_{>k}} a_n = 1.$$

In particular all square-free products compress exactly. For the coordinate-square defect, parity and the same branch calculation give

$$(\Phi(x_n^2) - I)\chi_F = \left(\prod_{i\in F_{>n}} a_i\right) \begin{cases} b_n - 1, & n \notin F, \\ R_n^2/b_n - 1, & n \in F \end{cases} \chi_F. \tag{B.1}$$

Theorem B.1. *Assume $\sum_n (b_n - 1) < \infty$. Let $w_n \in \mathbb{R}^D$ satisfy $\sum_n R_n|w_n| < \infty$, with at least one $w_n \neq 0$, and put*

$$t = \sum_n w_n x_n, \qquad s = \sum_n w_n \varepsilon_n.$$

There is no compact convex body with compact extreme boundary carrying an affine-agreeing pair (π, Ψ) for which the first D ambient coordinate functions τ_j satisfy $\pi(\tau_j) = M_{s_j}$ and

$$\Psi(\tau_j^2) \geq \Phi(t_j^2) \qquad (1 \leq j \leq D).$$

In particular, no positive barycentric-kernel construction realizes these data, regardless of its additional coordinates.

Proof. The product $a_* := \prod_n a_n$ is positive. Equation (B.1) gives

$$\Phi(x_n^2) - I \geq a_* \min\{b_n - 1, R_n^2/b_n - 1\} I > 0.$$

Uniform convergence and exact compression of mixed products imply

$$\Phi(t_j^2) - M_{s_j}^2 = \sum_n (w_n)_j^2 (\Phi(x_n^2) - I).$$

For some j, a nonzero summand bounds this defect below by a positive multiple of I. The asserted inequality for Ψ contradicts Lemma 2.3. For a barycentric kernel it follows from scalar Jensen's inequality for a coordinate square and positivity of compression. □

The same obstruction applies to finite nonlinear coordinates.

Proposition B.2. *Under the preceding summability hypothesis, every nonconstant real multiaffine polynomial p in finitely many coordinates satisfies*

$$\Phi(p^2) - M_{p(\varepsilon)}^2 \geq c_p I$$

for some $c_p > 0$. It therefore cannot be used as an affine coordinate of a compact-extreme-boundary barycentric construction.

Proof. Exact square-free compression identifies the defect with T^*T, where $T = M_p V - V M_{p(\varepsilon)}$. Restrict this error to the initial branch Y_0. If p involves the first N variables, its tail factor is bounded below by the positive product $\prod_{n>N} b_n^{-1/2}$. It suffices to prove injectivity of the finite-prefix error.

Identify functions on the binary cube with multiaffine polynomials. Let $\mathcal{D}$ scale a monomial x_F by $\prod_{i \in F} b_i^{-1}$. If the error vanishes on a vector represented by h, then, with $q = \mathcal{D}h$,

$$p(x)q(x) = \mathcal{D}(p(\varepsilon)h(\varepsilon))(x)$$

on the full three-point product grid. Both sides have degree at most two in each variable, so this is a polynomial identity. The right side is multiaffine. Whenever $p = p_0 + x_i p_1$ with $p_1 \neq 0$, write $q = q_0 + x_i q_1$; comparison of the x_i^2 coefficient gives $p_1 q_1 = 0$, hence $q_1 = 0$. Thus q and h are independent of every active variable of p. The identity reduces to

$$[p(x) - p(x_1/b_1, \ldots, x_N/b_N)]\, q(x) = 0.$$

The first factor is nonzero, because p is nonconstant and every $b_i > 1$. Hence $q = 0$. The finite-prefix error has positive minimum singular value; tensoring with the bounded-below tail proves the claim. □

Appendix C. Correlated magnitudes and an expectation obstruction

Independence of the magnitudes is not essential to the Walsh isometry. This observation gives a distinct family of exact compression maps. A simple expectation test, however, excludes an entire range of its possible geometric realizations.

Fix $b > 1$ and $0 < \delta < b$. Let θ be uniform on the circle and let ξ_n be independent uniform signs, independent of θ. Put

$$x_n = \xi_n \sqrt{b + \delta \cos(4^n \theta)}. \tag{C.1}$$

These coordinates share their angle and are not independent. Nevertheless, if $S_n = x_n^2$, then for every finite index set and integers $0 \le k_n \le 3$,

$$\mathbb{E} \prod_n S_n^{k_n} = \prod_n \mathbb{E} S_n^{k_n}. \tag{C.2}$$

Indeed, a nonconstant term in the Fourier expansion has frequency $\sum_n j_n 4^n$ with $|j_n| \le 3$. Its largest nonzero term cannot be cancelled, since $4^m > 3 \sum_{n<m} 4^n$. Thus only products of constant Fourier coefficients survive. By contrast,

$$\mathbb{E}[(S_n - b)^4 (S_{n+1} - b)] = \delta^5/16 \neq 0,$$

as follows from the $\cos 4t$ coefficient in $\cos^4 t$.

On the initial branch use (C.1) at every coordinate; on branch Y_k use the binary prefix, the zero-valued star, and the same correlated law on its tail. The angle and all prefix and tail signs are independent. For compactness, use disjoint tagged copies of the binary symbols and of the compact tail intervals, even where their numerical values agree. Each fixed branch is a compact image of its angle and sign space; branches with stars tending to infinity accumulate only on the binary source. Thus the union is compact, Y_0 is clopen, and every numerical coordinate is continuous.

Proposition C.1. *With these branch laws and the same formula for V as in (6.1), using $a = b^{-1}$, the map V is an isometry and every square-free product compresses exactly. Moreover $V1 = \mathbf{1}_{Y_0}$ and*

$$(\Phi(x_n^2) - I)\chi_F = b^{-|F_{>n}|} \begin{cases} b - 1, & n \notin F, \\ \delta^2/(2b^2), & n \in F \end{cases} \chi_F. \tag{C.3}$$

In particular each coordinate-square defect is positive and injective.

Proof. Independent signs make different monomial labels orthogonal. Equation (C.2) gives the same normalized monomial norms as the independent law. A surviving square-free matrix coefficient uses only exponents zero and two, so the complete proof of Lemma 6.2 applies. For the square defect, $\mathbb{E} x_n^4 = b^2 + \delta^2/2$. If $n \in F$, the total pre-star branch weight is $b^{-|F_{>n}|-1}$ and the weighted second moment is $(b^2 + \delta^2/2)/b$; the star contributes zero. Subtracting I gives the second coefficient in (C.3). If $n \notin F$, the calculation uses the ordinary second moment b, giving the first. □

Theorem C.2 (An obstruction independent of the height correction)**.** *For the correlated model, let $w_n \in \mathbb{R}^D$ be a nonzero absolutely summable sequence and put*

$$t = \sum_n w_n x_n, \qquad s = \sum_n w_n \varepsilon_n, \qquad C = s(G).$$

Suppose a continuous height z satisfies

$$\Phi(z) = M_{|s|^2}, \qquad z(y) \ge |t(y)|^2 - \operatorname{dist}(t(y), C)^2 \quad (y \in Y_0).$$

Then

$$\frac{1}{2\pi}\int_0^{2\pi}\sqrt{b+\delta\cos\theta}\,d\theta\le 1. \tag{C.4}$$

No additive form for z, and no geometric decay of $|w_n|$, is assumed.

Proof. Let $L=\sum_n|w_n|^2>0$ and denote the mean in (C.4) by m. Exact height compression and $V1=\mathbf{1}_{Y_0}$ give $\mathbb{E}_{Y_0}z=L$, whereas independent signs give $\mathbb{E}_{Y_0}|t|^2=bL$. For each branch point, compare t with the source point having the same signs. Averaging first over signs and then over the angle gives

$$\mathbb{E}_{Y_0}\operatorname{dist}(t,C)^2\le\mathbb{E}\left|\sum_n w_n\xi_n\bigl(\sqrt{b+\delta\cos(4^n\theta)}-1\bigr)\right|^2=(b+1-2m)L.$$

Containment gives the reverse lower bound $(b-1)L$. Dividing by L proves $m\le 1$. Absolute uniform convergence justifies all expansions. □

Corollary C.3. *The distribution $b=6/5$, $\delta=1$ admits no such exact height in any finite horizontal dimension. Suppose a body K satisfies the geometric hypotheses of Theorem 8.3, contains every encoded point $(t(y),z(y))$, and a positive barycentric kernel realizes the affine-agreeing map on the signed-source representation. Then no finite strictly convex graph can repair this model when $D\le 3$.*

Proof. The two chords of the square-root function on $[b-\delta,b]$ and $[b,b+\delta]$ give

$$m\ge\sqrt{b}+\frac{\sqrt{b+\delta}+\sqrt{b-\delta}-2\sqrt{b}}{\pi}>\frac{\sqrt{b}+\sqrt{b+\delta}+\sqrt{b-\delta}}{3}.$$

The last inequality uses strict concavity and $\pi>3$. At the stated parameters, the three square roots exceed $109/100$, $148/100$, and $44/100$, respectively. Thus $m>301/300>1$. For the graph assertion, exact mixed products and (C.3) give an injective defect in any nonzero horizontal coordinate. A positive barycentric kernel preserves that lower bound, and the signed source has full spectral support. Theorem 8.3 forces the excluded quadratic envelope. □

More generally, the mean-radius condition forces $1<b<\pi^2/8$ and $\delta^2>4(b-1)$. For the first inequality, the mean is strictly decreasing in δ and its limit at $\delta=b$ is $2\sqrt{2b}/\pi$. For the second, its value is strictly larger than the endpoint chord average $(\sqrt{b+\delta}+\sqrt{b-\delta})/2$. These restrictions leave admissible pairs near $(1,1)$; they do not exclude the whole correlated family or establish containment for any remaining pair.

Appendix D. Restrictions on direct operator constructions

We record two short diagnostics that are independent of the Walsh construction. They explain why two natural direct constructions cannot supply the desired diffuse example. Neither is a universal dimension bound.

Proposition D.1. *Suppose an affine-agreeing POVM E satisfies $E(B)\le\nu(B)I$ for every Borel $B\subset X$, where ν is a finite positive scalar measure. Then the spectral support of π has at most $\lfloor\nu(X)\rfloor$ points and $E=P$. In particular, an operator-valued density with integrable operator norm cannot yield a diffuse counterexample.*

Proof. For a spectral-support point x, choose unit vectors $h_n\in P(B(x,1/n))H$. Their scalar measures for E are probabilities dominated by ν, with barycenters tending to x. Any weak limit is δ_x by extremality, and remains dominated by ν. Therefore $\nu(\{x\})\ge 1$. There are at most $\lfloor\nu(X)\rfloor$ such points, and Lemma 2.2 gives equality of the maps. For a positive density F relative to a scalar measure μ, take $d\nu=\|F\|\,d\mu$. □

This is an order-domination hypothesis, not merely absolute continuity with respect to a scalar control measure. It does not automatically apply to a correction of a spectral measure.

Proposition D.2. *Let $\rho : C(X) \to L^\infty(\mathbb{T})$ be a unital $*$-representation, and let P_+ be the orthogonal projection from $L^2(\mathbb{T})$ onto the scalar Hardy space H^2. The compression*

$$\Phi(f) = P_+ M_{\rho(f)}|_{H^2}$$

cannot be a nontrivial UCP map agreeing with a representation on $A(K)|_X$.

Proof. Let S be the unilateral shift on H^2 and let e_0 be its first basis vector. For self-adjoint Toeplitz operators A, B, put $u_A = S^*Ae_0$ and $u_B = S^*Be_0$. The identities $S^*AS = A$ and $SS^* = I - e_0e_0^*$ give

$$S^*[A, B]S - [A, B] = u_Au_B^* - u_Bu_A^*.$$

If A, B commute and $u_A \neq 0$, the two rank-one products agree. It follows that $u_B = \lambda u_A$ for some real λ. For a real symbol f, u_{T_f} consists of its strictly positive Fourier coefficients. The negative coefficients are their conjugates. Consequently, commuting real scalar Toeplitz symbols are all of the form $\lambda_j f + c_j$ for one real symbol f, unless all are constant.

In an affine-agreeing compression the compressed coordinates commute, because they equal the coordinates of the representation. Their joint spectral support therefore lies on an affine line, or is a singleton. A line meets $\operatorname{ex} K$ in at most two points: every interior point of its intersection segment with K is nonextreme. The represented support is thus finite, and Lemma 2.2 applies. □

In contrast, small norm and low Kraus rank do not themselves impose rigidity. Any counterexample $\Psi(f) = V^*\rho(f)V$ can be amplified on $H \oplus L \oplus L \oplus \cdots$ to $\widehat{\pi} = \pi \oplus \rho \oplus \rho \oplus \cdots$. The isometry

$$W(h, \ell_1, \ell_2, \ldots) = (0, Vh, \ell_1, \ell_2, \ldots)$$

satisfies $W^*\widehat{\pi}(f)W = \Psi(f) \oplus \rho(f) \oplus \rho(f) \oplus \cdots$. Mixing this compression with the identity gives a nontrivial two-Kraus channel fixing the affine coordinates and arbitrarily close to the identity. This is a reformulation of a counterexample, rather than an independent construction of one.

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